Concealed Questions (Frana 2021)

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1 Introduction: some key facts

A concealed question (CQ) is a nominal phrase that behaves semantically like an embedded question when used as the complement of certain verbs. Verbs like *know, reveal, find out, guess*, etc. can take CQs.

(1) a. Mary knows Joan's phone number.
a'. Mary knows *what Joan's phone number is*.
b. The X-ray revealed the gender of the baby.
b'. The X-ray revealed *what the gender of the baby was*.

Some questions Frana (2021) aims to answer:

1. How do nominal phrases (e.g., *Joan's phone number*) take on question-like meanings? Why do some definite descriptions allow CQ readings while others don't?
2. What are the core properties of CQs?
3. How should CQs be analyzed formally? Which approach better accounts for the data?
4. Can CQs extend beyond definite descriptions?

Not every nominal phrase can take on a CQ meaning. Unlike regular definite descriptions, CQs do not simply refer expressions but involve **intensional meaning**.

(2) a. Sam paid / figured out the exact amount of her debt.
b. The exact amount of Sam's debt is $2,000.
c. Sam paid / #figured out $2,000.

CQs occur with **epistemic verbs**, but not every epistemic verb that take nominal phrases can syntactically embedded questions and allow for CQ readings of their DP complements.

(3) a. Meg forgot the capital of Italy. (CQ meaning: *Meg forgot what the capital of Italy is*.)
b. *Meg believed *what the capital of Italy is*.
c. *Meg believed the capital of Italy.

d. Kim knows the governor of Alaska. (CQ meaning: Kim *knows who the governor was*.)
e. *Kim believed *who the governer of Alaska was*.
f. *Kim believed the governor of Alaska.

Even though CQs act like embedded questions, they are syntactically definite descriptions. Not every question-embedding predicate can select for CQ complements.

(4) a. I wonder *what answer he gave*.
a'. *I wonder the answer he gave.
b. John inquired *what the number of students in the class was*.
b'. *John inquired the number of students in the class.

Syntactic vs. semantic selection?

(5) (from Frana 2021 (29))

	Syntactic subcategorization	*Semantic selection*
Know	[DP,CP]	[**Question**, Proposition]
Believe	[DP,CP]	[Proposition]
Wonder	[CP]	[**Question**]

2 Main CQ analyses

2.1 The individual concept (IC) approach

The IC approach, first proposed by Heim (1979), suggests that a CQ (*the capital of Italy*) denotes an **individual concept**, which is a function that maps possible worlds to individuals → Verbs like *know* and *figure out* select individual concepts rather than simple entities.

(6) a. The mayor of Amherst is Ms Higgins.
$\mathbf{f(i_0) = g(i_0)}$ *extensional identity*
b. The mayor of Amherst lives on Main St.
$\mathbf{lives\text{-}on\text{-}MS_{i0}\ (f(i_0))}$ *extensional predication*
c. Ms Higgins lives on Main St.
$\mathbf{lives\text{-}on\text{-}MS_{i0}\ (g(i_0))}$

(7) a. The capital of Italy is the largest town in Italy.
$\mathbf{f(i_0) = g(i_0)}$ *extensional identity*
b. John knows the capital of Italy.
$\mathbf{know_{i0}\ (f)(john)}$ *intensional predication*
c. John knows the largest town in Italy.
$\mathbf{know_{i0}\ (g)(john)}$

Romero (2005) proposes a **cross-categorical** analysis of the epistemic *know* so that the predicate can combine with both fully embedded questions (CPs) and concealed questions (individual concepts).

(8) $[\![know_{Q}]\!]^{w} = \lambda Q_{<s,<<s,t>,t>>}\ \lambda x_{e}.\ \forall w' \in \mathrm{Dox}_{x}(w)\ [Q(w') = Q(w)]$

(9) $[\![know_{CQ}]\!]^{w} = \lambda f_{<s,e>}\ \lambda x_{e}.\ \forall w' \in \mathrm{Dox}_{x}(w)\ [f(w') = f(w)]$

(10) $[\![\text{John knows}_{CQ}\text{ the capital of Italy}]\!]^{w} =$
$\forall w' \in \mathrm{Dox}_{John}(w)\ [\iota x_{e}\ [x \text{ is Italy's capital in } w'] = \iota x_{e}\ [x \text{ is Italy's capital in } w]]$

The IC approach avoids the need to assume an underlying *wh*-question structure. It also captures the difference between CQs (intensional objects) and extensional definite descriptions.

2.2 The clausal approach

The clausal approach, proposed by Nathan (2006), assumes that there is **a type-shifting mechanism** converts a NP into an identity question meaning.

A **type-shifting operator (Q-shifter)** is used to convert a definite noun phrase into a question meaning**.**

(11) Q shifter (after Nathan 2006 (6))
$[\![\mathbf{Q}]\!]^{w} = \lambda f_{<s,e>}\ \lambda p_{<s,t>}\ .\ [\exists x_{e}\ .\ p = \lambda w'.\ [f(w') = x]]$

(12) a. $\lambda p_{<s,t>}\ .\ [\exists x_{e}\ .\ p = (\lambda w'.\ [\![\text{the capital of Italy}]\!]^{w'} = x)]$
b. {the capital of Italy is Rome, the capital of Italy is Venice, the capital of Italy is Salerno, …}

(13) $[\![know_{Q}]\!]^{w} = \lambda Q_{<<s,t>,t>}\ \lambda x_{e}.\ \forall p\colon (Q(p)\ \&\ p(w)) \Rightarrow \forall w' \in \mathrm{Dox}_{x}(w)\colon p(w')$

(14) $[\![\textit{John knows}_{Q}\ \mathbf{Q}\ \textit{(the capital of Italy)}]\!]^{w} =$
$\forall p\colon ([\exists x_{e}\ .\ p = (\lambda w'.\ [\![\text{the capital of Italy}]\!]^{w'} = x)]\ \&\ p(w)) \Rightarrow \forall w' \in \mathrm{Dox}_{John}(w)\colon p(w')$

The clausal approach explains why CQs behave like embedded questions even though they are syntactically definite descriptions.

3 A more complex case of CQs: nested CQs

Heim (1979) discovered the ambiguity in sentences with **nested CQs** (a CQ that contains another CQ inside it, typically embedded in a relative clause).

(15) (Heim 1979 (34))
John knows the price that Fred knows.

Nested CQs allow both a question (Q) reading and a meta-question (MQ) reading.

(16) Q: John, like Fred, knows that milk costs $2.00 (per gallon).
MQ: John knows which price Fred knows. It is the price of milk.

The IC approach can only account for the Q reading. Romero (2005) provides a type-flexible *know*$_{CQ}$.

(17) $[\![\text{know}_{CQ}]\!]^w = \lambda \underline{y_\tau}\, \lambda x_e\, \forall w' \in \text{Dox}_x(w)\, [y(w') = y(w)]$

where τ = <s,e> or <s,<s,e>> or ...

(18) Q: $[\![\text{know}_{CQ}]\!]^w + [\![\text{the price that Fred knows}_{<s,e>}]\!]^w$

MQ: $[\![\text{know}_{CQ}]\!]^w + \lambda w'.\, [\![\text{the price that Fred knows}_{<s,e>}]\!]^{w'}$

Roelofsen and Aloni (2008) points out that Heim's sentences with nested CQs can ave two additional (set-based) readings.

(19) (Suppose that all that Fred knows is that $2.00 is the price of something in the list of products (butter, milk, cheese, etc.))

Set/Q: John, like Fred, knows that $2.00 is the price of some product.

Set/MQ: John knows that Fred knows the price of some product. It ís $2.00.

Another scenario that I can think of is that there could be a case where both milk and butter costs $2.00. John only knows milk costs $2.00, and Fred only knows butter costs $2.00. Since they both know $2.00 is the price of some product (either milk or butter), can the sentence *John knows the price that Fred knows* express this reading?

4 References

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